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Multi-Modal Biometric Authentication Using Attention based RNN Feature Extraction and Walrus-Assisted Sculptor Optimized Fusion

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Abstract— **This paper, proposes a novel framework for multimodal biometric authentication using face, fingerprint and iris images leveraging the efficiency of hybrid deep learning algorithms and meta-heuristic optimization for generating fused feature vectors for enhanced authentication. An Attention based RNN (AbRNN), with MobileNetV2 feature extractor for capturing rich and discriminative features from different modalities is used. A new hybrid metaheuristic, called Walrus assisted Sculptor Optimization Algorithm (WaSOA), is subsequently proposed, and on top of that employed to fuse them optimally by tackling the multimodal fusion with an intricate search space. The resulting fused and optimized feature vector is then fed into SVM classifier for robust authentication. The proposed feature extraction and fusion pipeline is validated on a multi-modal dataset and experimental results favor the proposed method in forming strong biometric features, which can be effectively utilized for improving biometric system performance and security. The designed framework was implemented in Python and validated using publicly available datasets. The experimental results are assessed and evaluated through comparative assessment in terms of metrics such as accuracy (99.50%), precision (99.51%), recall (99.50%), F1-score (99.5%), average FAR of 0.06%, average FRR of 0.50% with an average EER of 0.04%.**

Keywords- Multimodal Biometrics Recognition, Attention-based Recurrent Neural Network (AbRNN), Biometric fusion optimization metaheuristic.

I. INTRODUCTION

In today's digitally connected world, the growing need for reliable and practical identity verification has made biometrics—such as face, fingerprint, and iris—a vital component of contemporary security [1]. Even though multi-modal biometric systems are more secure and resilient to the drawbacks of single-modal methods [2], they pose considerable difficulties in efficiently extracting and cleverly combining heterogeneous features to produce discriminative feature vectors. Innovative architectural solutions that can generate highly consistent within-class and distinct between-class representations are required due to the intrinsic diversity of biometric data, the need for efficient feature extraction that is resilient to real-world variations, and the difficulty of optimal feature fusion [3].

An innovative architecture that tackles these issues is presented in this paper. The contributions include a novel Attention-based Recurrent Neural Network (AbRNN) for robust feature extraction from various biometric modalities and a novel Walrus assisted Sculptor Optimization Algorithm (WaSOA) for intelligently combining these heterogeneous features through adaptive learning of salient features and optimization of their combination, the suggested system achieves superior performance, opening the door for more dependable and secure biometric applications.

The rest of the paper is structured as follows: Section 2 provides a review of developments in the field of multi-modal biometric feature extraction and fusion techniques. Section 3 delves into the proposed architecture, explaining the design and rationale behind the Attention-based Recurrent Neural Network and the Walrus assisted Sculptor Optimization Algorithm. Section 4 presents and discusses the experimental results, showcasing the effectiveness of the proposed methods. Finally, Section 5 concludes the paper and suggests potential future research directions.

II. RELATED WORK

The integration of deep learning techniques has led to significant breakthroughs in the field of multi-modal biometrics. This section briefly reviews the key developments in the areas of feature extraction and feature fusion methodologies, laying the groundwork for the proposed architecture.

Advancements in the field of deep learning have greatly accelerated the development of biometric systems, especially multi-modal approaches. Mehdi et al. (2020) [4] proposes a CNN based biometric authentication of fingerprint, face and finger vein recognition, highlighting the widespread use of Convolutional Neural Networks (CNNs) in feature extraction

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process across various modalities owing to their ability to learn hierarchical data representations.

However, traditional CNNs find it difficult to model long-range dependencies or efficiently processing diverse, sequential image features. Ling et al. (2020) [5] came up with the idea of attention-based convolutional neural network (ACNN) for generation of discriminative facial feature embeddings. Suvidha et al. (2021) [6] showed how Bi-LSTM RNNs with attention are useful for getting features from image-derived sequences.

S. Maneet et al.et al. (2019) [7] gives a detailed review on multimodal fusion techniques and different levels of fusion (feature, score, decision), outlining their advantages and disadvantages. According to D. Cherifi et al. (2015) [8], methods such as Genetic methods (GA) and Particle Swarm Optimization (PSO) have generally been used to optimize this fusion, particularly at the feature level, in order to achieve discriminative authentication. This trend includes algorithms like the Walrus Optimization Algorithm (WOA) [9] and the Sculptor Optimization Algorithm (SOA) [10], which provide new ways to optimize fusion techniques.

Despite these developments, a number of research gaps remain. Current feature extraction techniques often lack explicit mechanisms to effectively process various image data sequentially or simulate long-range connections, especially for heterogeneous multi-modal inputs. Furthermore, current fusion techniques, whether classical or rely on well-established metaheuristics, often struggle to traverse the large, high-dimensional search space required for dynamic and intelligent weighting of various biometric information. This highlights the need for an integrated architecture that combines state-of-the-art attention-based RNNs for feature extraction with WaSOA metaheuristics for feature-level fusion. The proposed study introduces such a thorough and logical framework that specifically addresses these issues.

III. METHODOLOGY

This section explains in detail about the novel architecture designed to generate robust and highly discriminative multi-modal biometric fused feature vector. The system integrates an Attention-based Recurrent Neural Network (AbRNN) for feature extraction, a novel Walrus assisted Sculptor Optimization Algorithm (WaSOA) for intelligent feature fusion followed by a machine learning classifier.

A. System Architecture Overview

The proposed multi-modal biometric feature fusion architecture is illustrated in the block diagram given in Fig.1. The process begins with preprocessing of raw biometric inputs (face, fingerprint, and iris) obtained from dataset repositories [11,12,13], for removing noise followed by a series of feature extraction, fusion and optimization steps resulting in the generation of discriminative biometric embeddings suitable for security applications.

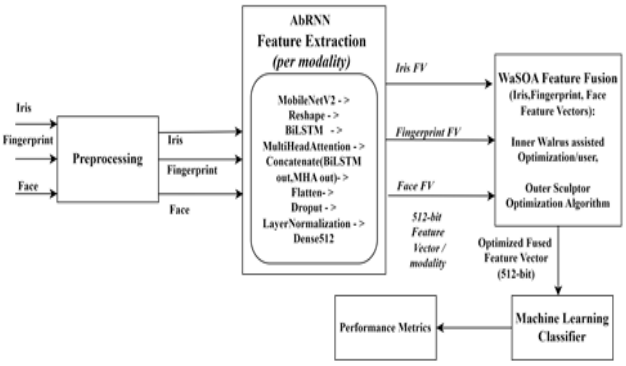


Figure 1. System architecture- overview

B. Pre-processing Biometric Data

In the pre-processing step, face, finger print and iris biometric data were first resized to 255 x 255 dimension for standardization, followed by Gaussian Blur (for noise reduction, resulting in image smoothening keeping key features intact) and Normalization (to [0,1] range) to ensure underlying embeddings depend on actual biometric features only.

C. Attention- based Recurrent Neural Network for Feature Extraction

The attention-based Recurrent Neural Network (AbRNN) was applied to improve the feature extraction process by integrating the efficiency of CNN [5], attention models [6] and spatial extraction of RNN [7] for input pre-processed images. Initially, the pre-processed biometric images for face, fingerprint and iris modality standardized to a size of 224x224x3, were represented as given in (1).

$$\bar{Y} \in \Re^{F \times FP \times I} \tag{1}$$

where, $\bar{Y}$ is denoted as parameter representation of biometric images, *F, FP* and *I* are expressed as pre-processed data based on face, fingerprint and iris biometric inputs. The core of AbRNN for each modality begins with a Convolutional Neural Network (CNN) backbone. The base feature extractor used here is ImageNet dataset pretrained MobileNetV2 model. MobileNetV2 is chosen because of its efficient architecture, providing robust, high-level feature extraction while maintaining computational lightness. The first 100 layers of the MobileNetV2 model are frozen (set to non-trainable) with the aim of utilizing its pre-learned generic image features and also prevent overfitting on potentially smaller biometric datasets. The output of CNN backbone after the processing of input image, is a feature map. Equation (2) given below represents the general operation of a convolutional layer, including activation and pooling.

$$F(l) = ReLU\left(W_l^{'} * F_{l-1} + b_l^{'}\right) \tag{2}$$

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where, $W_l^{'}$ denoted as convolution layer l with respect to the filter weights, $F_{l-1}$is represented as a feature map function with previous layer, b_l' is denoted as bias function. After applying these operations, the non-linearity of inputs is captured through a weighted filter contribution as given in (3).

$$Y(ij)_l = \sum_{m=0}^{a-1} \sum_{n=0}^{b-1} \sigma[x(ij)_l].W'_{mn} \quad (3)$$

where, m and n denote the spatial dimensions for calculating the pre-non-linearity function $\sigma[x(ij)_l]$. The CNN feature output corresponding to the MobileNetV2 backbone, has a shape of (7, 7, 1280). This 2D feature map is then reshaped into a 1D sequence of spatial features, denoted as $F' = [f_1', f_2', f_3', f_4', \ldots\ldots f_n']$, to facilitate sequential processing by the recurrent neural network layer, like a Bidirectional Long Short-Term Memory (Bi-LSTM) with 256 units. The Bi-LSTM processes the 1D spatial feature sequence in both forward and backward directions, thereby fully capturing the dependencies and relationships among the spatial features. The Bi-LSTM is configured to generate a sequence of hidden states, thereby enabling the application of attention mechanism dynamically across all temporal steps of the output. The hidden state function step of the RNN is generally given by (4).

$$Hs_t' = RNN_{Cell}(f_t', Hs_{t-1}') \quad (4)$$

The LSTM cells learn how features are related over time or space by managing their internal memory using various gates. Following the Bi-LSTM, a Multi-Head-Attention layer is applied to the output, which acts as a selective filter. Using multiple heads for attention, facilitates the model to dynamically determine and concentrate only on prominent spatial features within the Bi-LSTM's output, thereby enhancing the model's ability to extract and emphasize the truly discriminative regions which are relevant to biometric identification. The attention score computation is generally represented as given in (5).

$$A_t(l) = Score(H_t', z) \quad (5)$$

where, z is the hidden state query vector. Then to combine both contextual and attention-weighted features, the output of the Bi-LSTM and the attention layer are concatenated, followed by flattening to a single vector. To promote stable training and prevent overfitting, a Dropout (0.3) layer which is followed by a LayerNormalization layer is used here. The final Dense layer, based on *ReLU* activation outputs the final 512-dimensional feature vector for each pre-processed biometric image. This comprehensive integration of CNN, Bi-LSTM, and Multi-Head-Attention forms a robust AbRNN for discriminative feature extraction.

D. Walrus assisted Sculptor Optimization Algorithm for Feature Fusion

In this step, the AbRNN extracted feature vectors corresponding to Iris, Face and Fingerprint of a single user sample are fused using a novel Walrus assisted Sculptor Optimization Algorithm (WaSOA). WaSOA technique combines *local sample fusion* with *global modality weighting* using a two- level optimization strategy: an inner Walrus Optimization Algorithm (WOA) [9] and an outer hybrid fusion strategy based on Sculptor Optimization Algorithm (SOA) [10]. This two-level approach results in a fusion function, which is very much effective. The 512-dimensional feature vectors from feature extraction step, corresponding to each biometric instance of face, fingerprint, and iris modalities are conceptually summarized in the following equations (6) - (9).

$$Fb_{sequence} = [Fb_f, Fb_{FP}, Fb_I] \quad (6)$$
$$Fb_f = [f_1', f_2', f_3' \ldots . f_n'] \quad (7)$$
$$Fb_{FP} = [fp_1, fp_2, fp_3 \ldots\ldots\ldots fp_n] \quad (8)$$
$$Fb_I = [i_1, , i_2, i_3 \ldots . i_n] \quad (9)$$

where Fb_f, Fb_{FP} *and* Fb_I are denoted as the set of 512-dimensional feature vectors for face, fingerprint, and iris modalities respectively, for a given user.

The inner walrus optimization algorithm (WOA) [9], inspired by the behaviour of walruses is used to locally fuse multiple 512-dimensional feature vectors corresponding to the *same user within a single biometric modality* into a stable representative feature vector. As part of initialization, it generates a population of candidate weight vectors each representing a walrus. These weights are initialized randomly (Dirichlet distribution is used to guarantee weights add to 1). Equation (10) represents the objective function to assess the fitness of each walrus (candidate weight vector).

$$fitness(weights_{um}) = \frac{|fv_{um}|}{(1+np.\ std(fv_{um})+\varepsilon)} \quad (10)$$

The aim of *fitness (weights$_{um}$)* function is to maximize the magnitude or Euclidean norm of the resulting fused vector fv_{um} while minimizing its internal standard deviation thereby resulting in the generation of a robust, stable, and compact representative feature vectors for each user-modality pair. In the next step, inspired based on a set of walrus behaviours, iterative revision of WOA population (i.e. Weight vectors) takes place. The first behaviour is the *leader-follower dynamics*, in which agents adjust their positions or weights based on their target food (best known solution) and the best solution discovered so far (leader). The second is the *social component behaviour*, which mimics social interaction by allowing walruses to modify their movements according to the locations of other walruses in the population. Also, it uses the *Exploration/Exploitation behaviour*, which integrates additional escape mechanisms and foraging techniques that use random numbers and possibly moving in or out of the leader/food space in order to balance search space exploration with exploitation of promising locations. The search for the best local fusion weights is guided by these patterns of motion. The result is

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the generation of optimal weights (ow_{um}) corresponding to a user-modality pair, so as to fuse multiple feature vectors of a user(U)-modality(M) pair into a 512-dimensional representative fused vector, $RFV_{U,M}$. As part of achieving the global optimization goal, inner WOA also compute AV_{intra}, the average intra-class variability over all user-modality combinations. The outer layer of WaSOA uses Sculptor Optimization (SOA) [10] inspired from human sculpting process, for optimizing the global modality weights $(GW_{global} = [gw_I, gw_F, gw_{Fp}])$, using the representative feature vectors $RFV_{U,M}$ obtained using inner WOA. For optimizing the overall discriminative power of multi-modal system, this layer decides how to mix the representative feature vectors from various modalities. The process here involves building user-specific fused representatives across the three biometric modalities and computes the Global Objective function G_{ob} to assess the quality of a given collection of modality weights given in (11).

$$G_{Ob} = \frac{AV_{inter}}{(AV_{intra} + \varepsilon)} \quad (11)$$

The aim here is to maximize the average intra-class variability (AV_{intra}), obtained from the inner WOA layer, in relation to the average inter-class distance (AV_{inter}), which is the distance between various user's fused representatives, computed in this layer, resulting in maximizing the distance between distinct identities. By combining an exploratory search with a precise refinement, the Optimization Strategy coordinates the search for the ideal GW_{global}.

In Explorative Trial Generation phase, an initial set of trials with random global modality weight $GW_{global\ (j)}$) are generated (uses a Dirichlet distribution), and assessed using the global objective function. This *exploration* helps in identifying a set of promising regions within the solution space. Sculptor Optimization Algorithm (SOA) [11] inspired fine refinement is then performed on the trials found promising for a predefined number of iterations. This fine tuning of each modality weight uses the concept of *exploitation*, similar to a sculptor who carefully refines his work by making small precise changes. Here a Gaussian perturbation is used in each iteration, to change $GW_{current}$, the current best modality weights to $GW_{perturbed}$ as shown in (12). $\boldsymbol{G}W_{current}$ is changed to $GW_{perturbed}$ if Objective ($GW_{perturbed}$) is better than Objective ($GW_{current}$), finally resulting in identifying the optimal weights for each modality ($GW_{optimal}$).

$$GW_{perturbed} = GW_{current} + \boldsymbol{\varepsilon} \quad (12)$$

where $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I})$ is a vector of independent Gaussian (normal) random variables with mean 0 and standard deviation $\sigma = step_size$, which is a very small value and $\boldsymbol{I}$ is the identity matrix. These optimal weights, GW_{op}, are applied to the representative feature vectors ($RFV_{U,M}$) from each modality, resulting in the generation of F_{fused}, the final fused 512-dimensional feature vector for every biometric instance. Equations (13) and (14) represents the final fused output.

$$F_{fused} = Fuse_{F,W}(RFV_{U,I}, RFV_{U,F}, RFV_{U,Fp}, GW_{op}) \quad (13)$$

$$F_{fused} = F_1, F_2, \ldots, F_N \quad 14)$$

The generated 512-dimensional F_{fused} vector is stored and used as the input for SVM machine learning classifier.

IV. Results and Discussion

This section presents the experimental results of the proposed multi-modal biometric authentication based on novel AbRNN feature extraction and WaSOA feature fusion techniques and discusses its performance in comparison with other state of the art methods. Benchmark datasets were used for evaluating the performance of the proposed method, whose details are given in Table I. Data augmentation is used to increase the number of samples per user to ensure uniformity and improve learning.

TABLE I. Biometric dataset description

Name	Details
Fingerprint (FVC 2000) [11]	3520 images in 4 databases each having 110 unique finger impressions with 8 impressions /finger with different resolutions
Face (CASIA Face V5) [12]	2500 images of 500 subjects with 640 x 480 resolution
IRIS (CASIA IrisV2) [13]	2400 images of 120 individuals with 20 samples/individual and 640 x 480 resolution

The developed framework was implemented in Python language and experimental results were determined in terms of parameters such as accuracy, precision, etc. using SVM classifier. The developed model has a higher accuracy of 99.50%, precision of 99.51%, recall of 99.50%, F1-score of 99.50%, average FAR of 0.06%, average FRR of 0.50% with an average EER of 0.04%. The performance metrics plot of the proposed method along with ROC and DET curves is given in Fig.2 (a) -2(c).

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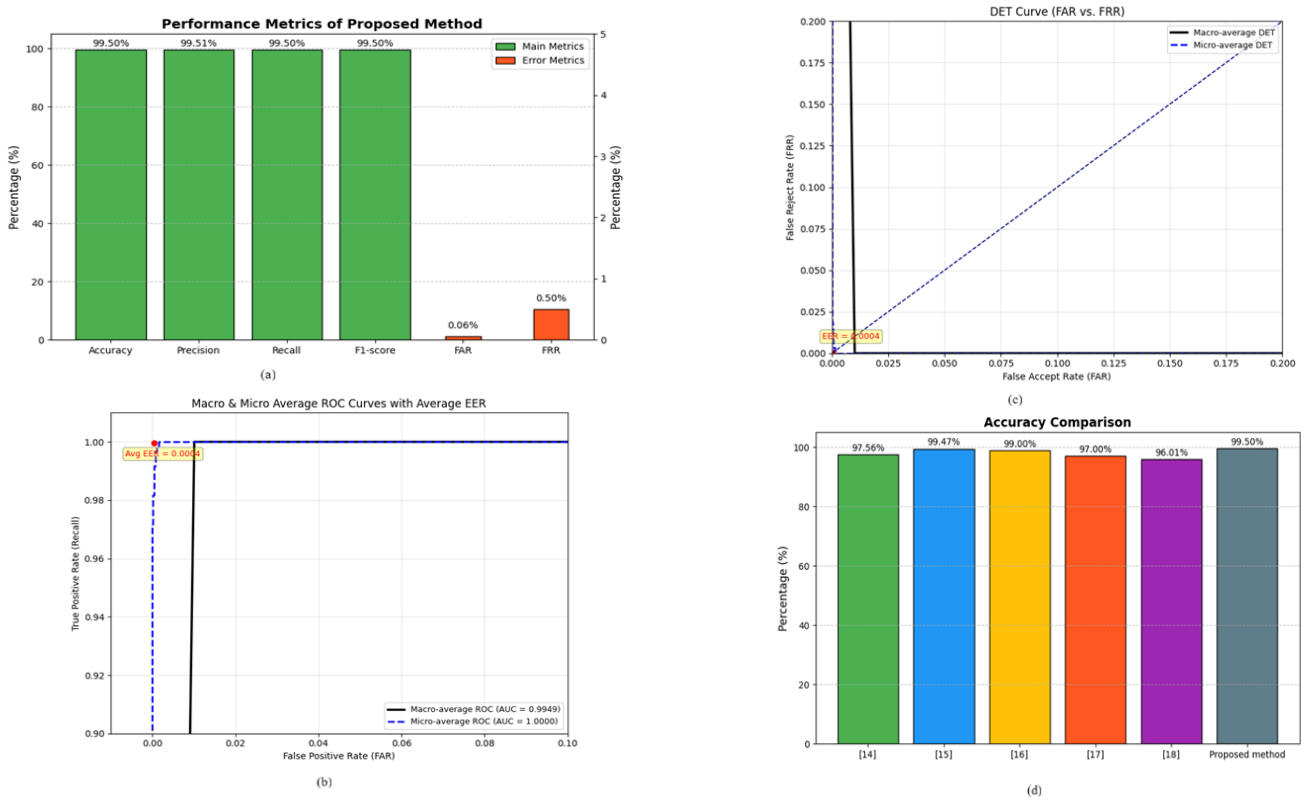


Figure 2. Performance plots of proposed method (a) performance metrics (b) ROC curve (c) DET curve (d) accuracy comparion with other techniques

To analyse the effectiveness of Attention based RNN feature extraction in identifying and focusing on relevant regions, attention heatmaps given in Fig.3 was generated from face, fingerprint and iris sample images. These heatmaps are a clear demonstration of the discriminative ability of attention mechanism within the RNN, supporting its use in extracting rich, contextual, and modality-specific features.

The multi-modal biometric authentication system, combining Attention-RNN feature extraction and WASOA-optimized fusion outperforms other methods in performance. A comparative analysis against prominent state-of-the-art multi-modal biometric authentication methods is presented in Table II. The proposed framework consistently outperforms these existing methods, particularly in terms of accuracy and simplicity, affirming its superior ability to handle the complexities of multi-modal biometric data. The results distinctly indicate that WASOA consistently achieves more discriminative fused feature vectors, leading to a significant improvement in authentication metrics. This enhancement is attributed to WASOA's unique hybrid search mechanism, which effectively navigates the complex feature space to determine optimal fusion parameters (e.g., weights for individual features or modalities). The accuracy-based performance comparison of proposed method with other multimodal feature extraction and feature fusion-based authentication techniques is given in Fig. 2(d). Though [15] claims around 99.47% accuracy value, it's feature extraction techniques are different for each biometric modality. Also, it uses a combination of all three feature fusion techniques to make a decision.

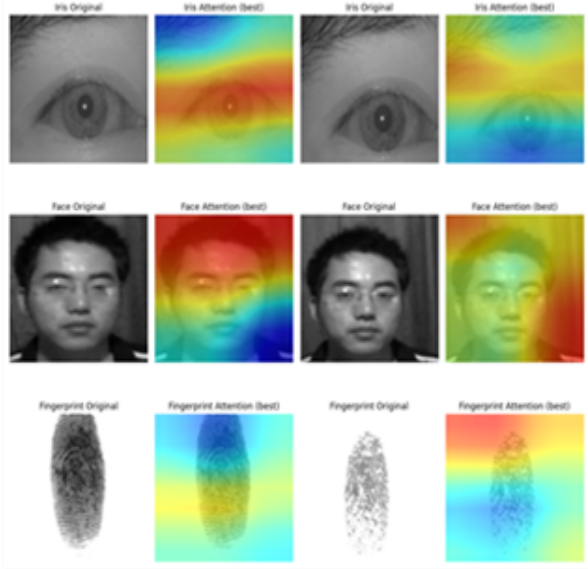


Figure 3. Iris, Face and Fingerprint Attention Heatmaps.

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TABLE II. Comparison of proposed method with existing methods based on feature extraction and feature optimization techniques.

Ref. No:	Modalities	Feature Extraction	Fusion & Authentication	Accuracy
[14]	Face, Finger-print, Iris	Gabor filter	SVM followed by RF using Bacterial Foraging Optimization (BFO) and Genetic Algorithm (GA) for feature level fusion	97.56%
[15]	Face, Finger-print, Iris	DL based Specialized feature extraction/ modality	Combines feature level fusion (PCA+GA), score level fusion (SVM enhanced min-max normalization and weighted sum rules) and decision level fusion (voting mechanisms +AND/OR rules)	99.47%
[16]	Finger-print, Iris, ECG	Swin Transformer + CNN	Feature-Level Fusion-scaled after concatenation of top 1000 features from (Swin transformer + CNN)/modality followed by Classification Head based authentication	99%
[17]	Face, Finger-print	Scale-Invariant Feature Transform (SIFT)	Feature-Level Fusion with Optimization using Binary chimp optimized adaptive kernel support vector machine (BCO-AKSVM)	97%
[18]	Voice, Finger-print, Iris	Multi-scale Residual Attention Network (RAN) with Atrous Spatial Pyramid Pooling (ASPP)	Feature level Fusion using Enhanced Lichtenberg Algorithm (ELA) optimization and Adaptive Recurrent Neural Network for authentication	96.01%
Proposed method	Face, Finger-print, Iris	Attention based RNN	Feature Level fusion using Walrus assisted Sculptor Optimization	99.50%

The Attention-RNN's adaptive feature extraction, combined with WASOA's intelligent fusion, creates a highly discriminative and robust fused feature representation. The SVM classifier then uses this optimized feature vector for precise authentication.

V. Conclusion and Future Scope

This paper introduces a novel multimodal biometric authentication using face, fingerprint, and iris, using Attention-based Recurrent Neural Network (AbRNN) for feature extraction followed by intelligent optimization of feature fusion process using Walrus assisted Sculptor Optimization Algorithm (WaSOA) model. Results show that Attention based RNN is effective in capturing complex and contextual feature representations utilizing the sequential processing power of RNNs combined with the ability of attention mechanisms to focus on salient regions, thereby enriching the discriminative power of the individual biometric features. Unlike other feature fusion methods, WaSOA's unique hybrid fusion approach intelligently optimizes the multimodal feature vectors resulting in a fused feature vector that is exceptionally robust and compact. Experimental findings using SVM classifier (99.50% accuracy) prove that there is significant improvement in authentication accuracy and robustness, confirming the effectiveness of the integrated approach in processing and combining complex biometric data for authentication applications. The robust and discriminative multi-modal biometric feature vectors generated by this framework can be used with further modifications for developing a secure biometric cryptosystem.

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